When Post-Editing Helps and When It Harms: Evaluating LLM Post-Editing of NMT for Tagalog and Spanish

Norris Research Project — Working Draft
California State University, Dominguez Hills
Research assistant: Shazan Ansar Mohammed; principal investigator: Benyamin Ahmadnia

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***Abstract*—Automatic post-editing can improve a machine translation draft while also introducing unnecessary or unsupported changes. We study this trade-off for English–Tagalog and English–Spanish in both directions, using a sentence-split NLLB-200 3.3B baseline and two post-editors: Llama 3 8B Instruct and GPT-4o Mini. Three strategies provide a source and draft alone, add one retrieved development exemplar, or require structured error diagnostics before editing. This development manuscript reports a completed pilot on the same 100 aligned FLORES+ development IDs, yielding 2,400 system outcomes. GPT retrieved one-shot improves English-to-Tagalog COMET by 1.463 points, with a descriptive paired 95% interval of [0.710, 2.396]. Llama reason-then-edit requires baseline fallback on 227 of 400 outcomes, compared with eight for GPT; these failures remain in the evaluation. Source-based inspection also identifies numeric corruption despite a metric gain and an apparently acceptable paraphrase with a substantial metric penalty. The pilot therefore motivates separate measurement of average quality, preservation, and output validity. It does not establish universal language-resource effects or intrinsic benefits of reasoning. Main devtest results, confirmatory comparisons, and final protocol decisions remain pending.**

***Index Terms*—automatic post-editing, machine translation, Tagalog, Spanish, retrieval, over-editing**

I. Introduction

An automatic post-editor must decide both what to correct and what to preserve. A fluent rewrite can repair an omission, but it can also alter a name, number, or source relationship that the original translation expressed correctly. For strong machine translation drafts, mean quality gains alone consequently give an incomplete account of editing behavior. Prior work demonstrates that large language models (LLMs) can improve neural machine translation (NMT), while also documenting hallucinated edits [1].

We investigate this tension using English–Tagalog and English–Spanish in both directions. A fixed NLLB-200 model [2] supplies the draft, and two instruction-following editors receive the source and draft. Comparing source-conditioned zero-shot editing, retrieved one-shot editing, and structured reason-then-edit separates practical prompting conditions without assuming that a diagnostic explanation is correct. The two language pairs provide a limited resource contrast; language, direction, domain, and starting quality remain intertwined.

The study asks four questions. **RQ1:** How do editing gains and harms differ between the Tagalog and Spanish pairs, conditional on baseline quality? **RQ2:** How do the three prompting strategies compare within each model and direction? **RQ3:** When do editors change already strong drafts and degrade their measured quality or source fidelity? **RQ4:** How do effects differ between the two directions of a language pair?

This manuscript develops an empirical evaluation, rather than a new post-editing architecture. Its current contributions are:

- a controlled four-direction pilot with two editors and three strategies, using the same aligned development sample;
- explicit accounting for schema validity, baseline fallback, textual editing, and quality change, with failed jobs retained;
- a preservation analysis that distinguishes metric-defined harm from source-supported factual errors and from heuristic flags.

The present evidence covers development only. A planned held-out experiment will address the research questions at the main-evaluation scale after the protocol is finalized. No devtest result is inferred from the pilot.

II. Related Work and Contribution Boundary

A. LLM post-editing and diagnostic feedback

Raunak et al. [1] evaluate GPT-4 as a direct translation post-editor and report both quality improvements and hallucinated edits. Ki and Carpuat [3] investigate external error annotations for LLM post-editing, including prompting and fine-tuning. Their findings motivate treating the usefulness of feedback as an empirical question. Our diagnostic condition instead asks an unmodified editor to produce its own short error categories. Those self-generated categories are observable outputs, not human annotations or verified explanations.

xTower [4] explicitly studies explanations and corrected translations, including expert assessment of explanation quality. Our experiment does not reproduce that trained system or its explanation evaluation. It asks whether a short diagnostic contract is operationally reliable for two fixed editors. LangMark [5] provides multilingual source–MT–human-post-edit triplets, including Spanish. FLORES+ instead supplies translation references, not human minimal post-edits. Our

reference-based gains therefore measure translation quality proxies and cannot establish reduced human editing effort.

B. Examples and multilingual representations

Vilar et al. [6] investigate translation prompting and example selection. We use LaBSE [7] to rank development sources by cosine similarity and supply one parallel demonstration. Similarity is semantic: the retrieval procedure does not independently measure syntactic similarity. A single deterministic exemplar makes the comparison reproducible, but does not establish that one exemplar or this encoder is optimal.

C. Relationship to the earlier ICAI study

The supplied ICAI manuscript [8] studies English-to-German domain adaptation with a frozen OPUS-MT draft, GPT-4 Turbo, multilingual Sentence-BERT top-three exemplars, and an optional second terminology-verification call. Frozen-NMT anchoring and dense exemplar retrieval are substantive shared methods. We cite these components as established, and do not claim the first hybrid NMT–LLM pipeline or first retrieval-based post-editor.

Norris changes the empirical setting: a stronger NLLB baseline, four Tagalog/Spanish directions, commercial and open-weight editors, and explicit analysis of strict output-contract failure and over-editing. Our reason-then-edit uses one generation call, rather than ICAI's second verification call. Different data, models, metrics, and controls preclude numerical cross-paper superiority claims. The exact publication metadata of the supplied ICAI version must be reconciled before submission; its title and authors are verified from the supplied paper.

D. Evaluation

BLEU [9], chrF/chrF++ [10], [11], and source-aware COMET [12], [13] offer complementary views. Reproducible lexical scoring requires full metric signatures [14]. Neither lexical overlap nor a neural scalar replaces source inspection. Our evaluation therefore reports measured quality changes alongside preservation screens and contract failures, without treating any screen as a gold error annotation.

III. Methodology

A. Frozen NMT baseline

The draft model is `facebook/nllb-200-3.3B`. The primary condition translates source sentence fragments independently and rejoins the target fragments with one space. The unsplit condition remains a diagnostic reference. Segmentation uses `pysbd` 0.3.4 with character spans and `clean=False`; Tagalog uses English rules as a documented heuristic. Dev-reviewed abbreviation and continuation safeguards accompany these rules, as recorded in Report 3's baseline evidence. Concatenation must recover the original source exactly before translation. This establishes lossless source segmentation, not preservation of cross-sentence context.

The completed development baseline uses FP16, four beams, no sampling, early stopping, length penalty 1.0, batch size 32, and maximum length 512. Inputs are checked for length rather than silently truncated. Single-fragment items reuse the control hypothesis. All 3,988 current unsplit hypotheses match the archived Phase 2 outputs. The archived generation specification and output hashes are available, although the original split-generation cell was not recovered in this review.

B. Editors and shared constraints

The editors are the original `Meta-Llama-3-8B-Instruct` checkpoint and the dated `gpt-4o-mini-2024-07-18` snapshot [15], [16]. Llama 3.1 is not substituted. Both receive the source, target language, and NMT draft. The shared instruction makes the source authoritative, requires preservation of source-supported names, numbers, dates, and meaning, and discourages stylistic rewriting. An accurate, natural, grammatical draft should be returned unchanged. Input strings are delimited as data.

C. Prompting strategies

Zero-shot returns one JSON field, `edited_translation`. **Retrieved one-shot** adds one solved development example, containing its source, NMT draft, and gold target, before the same query. Query references are absent. **Reason-then-edit** requests a diagnostic array followed by the final translation in a separate field. Diagnostics use six categories: mistranslation, omission, unsupported addition, morphosyntax mismatch, named-entity corruption, or no correction needed (`NONE`). Categories must be unique; `NONE` must occur alone; each note contains at most 25 whitespace-separated words. Full prompt bytes and schemas are retained in the candidate manifest.

D. Retrieval and leakage controls

LaBSE embeds the 997 development sources for each of three source languages into normalized 768-dimensional float32 vectors. Exact FAISS inner-product search ranks cosine similarity, with sentence-ID tie breaking. All 100 pilot IDs are excluded from every demonstration pool, leaving 897 candidate IDs before duplicate-source exclusions. Exact source duplicates are also excluded. The selected demonstration may expose its own development reference; the current query reference and quality scores never enter the prompt. All 400 pilot retrieval links were independently reconstructed from the saved vectors. The held-out retrieval pool policy must be recorded at freeze; it is not silently inferred from this pilot exclusion rule.

E. Structural validity and fallback

vLLM guided decoding and API structured output constrain JSON structure. Pydantic checks additional semantic contract rules after generation. Parsing success, schema validity, and usable completion are distinct. An invalid or unfinished response retains its raw record and returns the unchanged NMT draft for evaluation. Content failures are neither repaired nor selectively regenerated. The operative system output is

$$\hat{y}_i = \begin{cases} e_i, & \text{if the response passes the output contract,} \\ b_i, & \text{otherwise,} \end{cases} \tag{1}$$

where b_i is the baseline and e_i the extracted edit. Contract acceptance does not imply factual correctness. The evaluated condition includes this fallback policy.

IV. Data and Evaluation Protocol

A. Partitions and sampling

The experiment uses revision-pinned FLORES+ [2], [17], the maintained continuation of FLORES-200, with 997 development and 1,012 devtest items per direction. Its references are translations rather than human post-edits of our NMT drafts. The dataset's `fil_Latn` configuration maps to NLLB's `tgl_Latn`; English and Spanish use `eng_Latn` and `spa_Latn`. Spanish is the benchmark's standard variety, with no claim about heritage, dialectal, code-switched, or Los Angeles community Spanish.

A seed-42 uniform sample of 100 aligned development IDs is reused across all four directions, three strategies, and two editors. This gives 2,400 system outcomes, not 2,400 independent sentences. Conditions are balanced in count; the sample is not stratified by baseline quality. No pilot decision uses devtest scores. Older Phase 2 devtest baseline files are separate and unscored in this development workflow.

B. Metrics and uncertainty

The primary metric is reference-based `Unbabel/wmt22-comet-da` [18], reported as native score multiplied by 100. This identifies the exact checkpoint, rather than assuming the full ensemble from the COMET-22 shared-task submission. Secondary metrics use SacreBLEU 2.6.0: mixed-case BLEU with 13a tokenization and exponential smoothing; chrF++ with character order 6, word order 2, and $\beta = 2$.

For each condition, $d_i = C(x_i, \hat{y}_i, r_i) - C(x_i, b_i, r_i)$ uses native segment scores and the same source and reference. We report $100\bar{d}$ and a descriptive percentile interval from 10,000 paired bootstrap resamples, seed 42. The pilot reports no confirmatory p-values or multiplicity-adjusted significance claims. Separate baseline intervals are not tests of strategy superiority. Corpus BLEU and chrF++ are computed over all planned items, including fallback outputs.

C. Editing, harm, and preservation

Textual editing means $\hat{y}_i \neq b_i$, after the recorded outer-whitespace handling. Metric improvement is $d_i > 0.02$, harm is $d_i < -0.02$, and the inclusive middle band is $[-0.02, 0.02]$. A native difference of 0.02 equals two reported COMET points; the middle band is not textual identity or demonstrated equivalence.

The protect pool is fixed using original baseline scores $C_i \geq 0.90$, and the improve pool uses $C_i < 0.80$. Protect-pool over-editing requires a textual edit and metric harm. Its primary descriptive denominator is every protect item; the edited-protect denominator is also retained. These are metric-defined outcomes, not human judgments. Digit multisets, newly absent capitalized source tokens, target-language identification, and length ratios provide heuristic screens. The pilot used `langdetect` 1.0.9, not the fastText implementation named in the original roadmap. Legitimate localization and translation can trigger preservation flags.

TABLE I
Sentence-split baseline on the 100-ID development pilot. These are not the full 997-item development scores.

Direction	BLEU	chrF++	COMET
En→Tl	35.884	60.632	84.442
Tl→En	46.957	67.821	87.118
En→Es	27.694	54.086	86.076
Es→En	29.780	58.116	87.829

D. Execution and planned held-out study

Both editors use temperature 0, top_p 1, seed 42, and an output cap of 1,024 tokens. Llama runs in BF16 with vLLM 0.11.2 on an A100 80 GB, context limit 8,192. The GPT pilot uses four workers and four request starts per second. Exact revisions and environments appear in the appendix and manifest. Deterministic settings do not guarantee bit-identical remote service behavior.

The planned main matrix contains $1,012 \times 4 \times 2 \times 3 = 24,288$ outputs if the model/strategy matrix is retained. Before execution, the protocol must finalize retrieval eligibility, output policy, the complete contrast and multiplicity plan, and any documented deviations. The present manifest explicitly remains unfrozen. No held-out inference or confirmatory finding is claimed here.

V. Development Results

A. Segmentation diagnostic

Over all 997 development items per direction, sentence splitting changes mean COMET by +0.351 for English-to-Tagalog, +0.053 for Tagalog-to-English, +0.292 for English-to-Spanish, and +0.142 for Spanish-to-English. The Tagalog-to-English interval, [-0.019, +0.130], includes zero. The approved split baseline remains primary; these baseline segmentation changes are separate from post-editing gains.

Table I gives the baseline on the actual 100-ID pilot. Table II reports every post-editing condition, retaining all planned items and failures. The sample differs from the full 997-item baseline population.

B. Quality changes

GPT retrieved one-shot has the largest English-to-Tagalog point estimate: +1.463 COMET points, with descriptive interval [0.710, 2.396]. Both Tagalog directions have positive GPT zero-shot and one-shot point estimates. For Spanish-to-English, all three GPT intervals include zero. This evidence does not establish a general low-resource advantage: baseline quality, translation direction, and language pair are confounded.

Llama one-shot estimates are +0.087, +0.352, +0.450, and -0.046 points in the four directions ordered as English-to-Tagalog, Tagalog-to-English, English-to-Spanish, and Spanish-to-English. All four intervals include zero. Reason-then-edit

TABLE II

DEVELOPMENT PILOT RESULTS. EACH ROW USES ALL 100 PLANNED ITEMS, INCLUDING FALLBACK. COMET (C) AND DIFFERENCES (Δ) ARE REPORTED ON THE NATIVE $\times 100$ SCALE. INTERVALS ARE DESCRIPTIVE PAIRED 95% DEVELOPMENT INTERVALS; NO MAIN-EXPERIMENT SIGNIFICANCE IS CLAIMED. ZERO/ONE/REASON DENOTE ZERO-SHOT, RETRIEVED ONE-SHOT, AND REASON-THEN-EDIT.

Direction	Strategy	BLEU	chrF++	C	ΔC	95% dev interval	Fallback
GPT-4o Mini							
En→Tl	Zero	36.910	61.240	85.203	+0.761	[+0.147, +1.608]	0
En→Tl	One	37.757	62.087	85.905	+1.463	[+0.710, +2.396]	0
En→Tl	Reason	36.052	60.822	84.698	+0.255	[-0.726, +1.259]	3
Tl→En	Zero	47.144	68.470	87.988	+0.870	[+0.396, +1.425]	0
Tl→En	One	46.989	68.790	88.158	+1.039	[+0.471, +1.662]	0
Tl→En	Reason	47.408	68.467	87.736	+0.617	[+0.154, +1.169]	4
En→Es	Zero	28.672	54.627	86.741	+0.664	[+0.238, +1.216]	0
En→Es	One	28.260	54.475	86.695	+0.619	[-0.077, +1.295]	0
En→Es	Reason	27.946	54.180	86.077	+0.001	[-0.251, +0.298]	1
Es→En	Zero	30.297	58.418	87.846	+0.018	[-0.182, +0.194]	0
Es→En	One	29.821	58.671	88.131	+0.303	[-0.048, +0.698]	0
Es→En	Reason	29.486	57.846	87.601	-0.228	[-0.591, +0.084]	0
Llama 3 8B Instruct							
En→Tl	Zero	34.201	58.363	84.348	-0.094	[-1.158, +1.071]	0
En→Tl	One	33.799	59.280	84.529	+0.087	[-0.997, +1.303]	0
En→Tl	Reason	34.222	59.139	83.806	-0.637	[-1.405, +0.389]	39
Tl→En	Zero	45.269	66.959	87.131	+0.013	[-0.456, +0.500]	0
Tl→En	One	45.659	66.896	87.471	+0.352	[-0.115, +0.932]	0
Tl→En	Reason	45.224	66.686	86.574	-0.544	[-0.959, -0.217]	62
En→Es	Zero	27.125	53.821	86.431	+0.355	[-0.037, +0.797]	0
En→Es	One	27.381	53.815	86.527	+0.450	[-0.050, +0.979]	0
En→Es	Reason	26.558	53.267	85.864	-0.212	[-0.570, +0.153]	65
Es→En	Zero	29.776	57.955	87.636	-0.192	[-0.553, +0.128]	0
Es→En	One	29.942	57.590	87.783	-0.046	[-0.476, +0.384]	0
Es→En	Reason	29.202	57.240	87.267	-0.562	[-1.025, -0.192]	61

has negative Llama point estimates in every direction. Its intervals exclude zero for the two into-English directions, but remain exploratory. These effects describe the complete operational condition; they do not isolate reasoning from contract compliance.

C. Validity and failure behavior

All 800 zero-shot/one-shot responses from each model are usable. In reason-then-edit, only 173/400 Llama responses pass the contract, versus 392/400 GPT responses. Thus Llama uses baseline fallback on 227/400 outcomes (56.75%) and GPT on 8/400 (2%). Parsing alone is less informative: 399/400 Llama diagnostic responses parse as JSON.

Llama failures include repeated categories (184), invalid note length or emptiness (45), `NONE` combined with another category (23), diagnostic-count violations (3), and one unparseable, abnormally finished response. These screens overlap and must not be summed. GPT's eight failures all repeat a category. A separate GPT `NONE` response changes only hyphen spacing; it is disclosed as an instruction-consistency flag and is not retroactively reclassified.

D. Editing and protected drafts

Table III pools directions descriptively, with 400 outcomes per condition. Retrieved one-shot edits more often than zero-shot for both models. GPT one-shot yields more metric improvements but also more metric harms than GPT zero-shot. Low Llama reason-then-edit activity cannot be called successful restraint because most outputs use fallback. The protect denominator is 116 observations per strategy, with direction-specific counts 22, 45, 21, and 28. Only three Spanish-to-English items fall in the improve pool, limiting subgroup interpretation.

TABLE III

POOLED DESCRIPTIVE PILOT COUNTS PER 400 OUTCOMES. E: TEXT EDITED; I/H: METRIC IMPROVEMENT/HARM; P: PROTECT-POOL OVER-EDITS DIVIDED BY ALL 116 PROTECT OBSERVATIONS.

Editor	Strategy	E	I	H	P
GPT	Zero	143	38	8	3/116
GPT	One	320	67	19	6/116
GPT	Reason	97	26	20	3/116
Llama	Zero	260	37	32	9/116
Llama	One	326	49	37	11/116
Llama	Reason	158	8	48	16/116

Figure 1 complements mean differences by retaining the discrete mass at zero and the distribution tails. The curves are descriptive; no pooled independence assumption or stochastic-dominance test is implied.

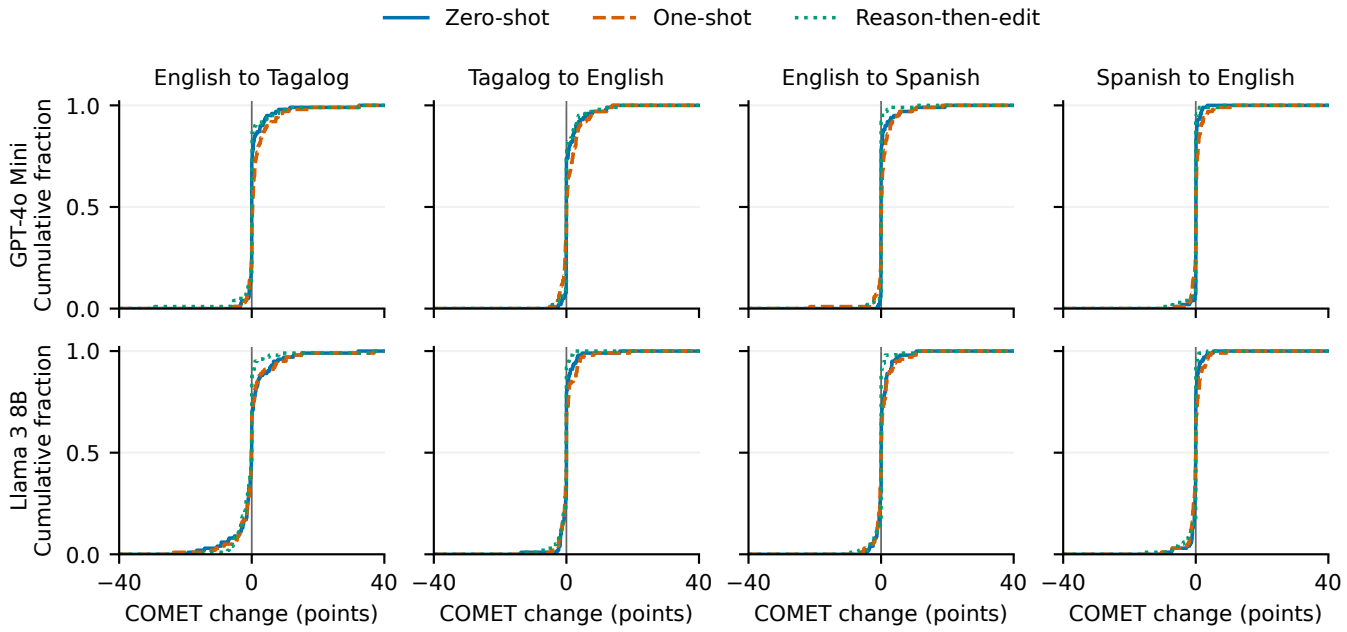


Fig. 1. Empirical distributions of development COMET changes, including every planned outcome. Each curve contains 100 items. The shared point scale retains all observations. Jumps at zero include unchanged translations and baseline fallback; especially for Llama reason-then-edit, that mass must not be interpreted as successful restraint.

E. Audit and resource accounting

The review independently reproduced all BLEU/chrF++ values from raw text, COMET means and paired intervals from aligned numeric caches, raw-record hashes, fallback decisions, and retrieval selections. Neural COMET inference was not rerun during this review. The supplied evaluator rechecked all 400 pilot baselines; the maximum difference was 5.96×10^{-7} native, below the 10^{-5} tolerance.

GPT used 643,126 input and 53,095 output tokens with no recorded retries. Applying the recorded uncached token rates gives an estimated charge of approximately $0.13, not an invoice. Six server fingerprint values occurred within the dated snapshot. Llama's 78 unique generation batches sum to 249.83 seconds of logged wall time, including reused smoke records; initialization is separate. GPT request wall time and Llama batch-amortized time are not comparable per-item speed measurements.

VI. Discussion and Source-Based Error Review

A. Validity is a measured part of the system

The diagnostic condition's contract is deliberately stricter than the plain translation schema. Its high Llama failure rate therefore concerns the joint model–prompt–validator system, rather than proving that explanations intrinsically impair translation. Filtering to accepted responses would change the sample in a model-dependent way. Retaining fallback outputs measures the deployable policy and prevents successful-response selection bias.

The development evidence supports disclosing this limitation before finalizing the protocol. Retaining the measured condition preserves the planned comparison; revising the diagnostic contract would require a documented change and affected-condition revalidation on dev. The current manuscript does not resolve that decision on behalf of the PI.

B. Average quality and fidelity can disagree

Source-based cases expose different failure mechanisms. In English-to-Tagalog item 429, Llama one-shot changes the source-supported area 23,764 to 27,874 square kilometres, yet COMET rises by 0.950 points. This is a factual corruption that a positive metric delta fails to identify. In Tagalog-to-English item 849, an accepted Llama diagnostic response changes the name Hangi Meal to The Last Meal and loses 13.998 points. Valid JSON and a confident error explanation do not establish fidelity.

Conversely, GPT one-shot's English-to-Spanish item 825 replaces *a bajo costo* with *de manera económica* in a source-consistent travel sentence, while COMET falls by 21.483 points. Inspection does not identify an obvious meaning loss commensurate with the penalty. We retain the metric flag and identify the need for linguistic confirmation; we do not label it a gold hallucination.

Preservation screens also have false positives. Three Llama digit-loss flags and both GPT digit-loss flags arise from correcting a duplicated baseline expression in item 296 from "100-foot-long [100 m]" to "100-meter-long." Transliterated names and valid numeral scaling trigger other screens. Zero language-mismatch flags do not imply zero semantic errors.

C. Influence and generalization

English-to-Tagalog item 983 distinguishes bird flu from the formal term avian influenza, whereas the baseline repeats the same Tagalog expression. GPT zero-shot and one-shot restore the distinction and gain 32.361 segment points. A post hoc leave-one-item-out diagnostic changes GPT one-shot's mean from +1.463 to +1.151 points and Llama one-shot's from +0.087 to -0.285. All 100 items remain in the primary results. This sensitivity discourages declaring an early strategy winner or adjusting held-out evaluation to favor the pilot.

VII. Ethical Considerations and Reproducibility

The study uses benchmark sentences and automatic evaluation without recruiting external raters. Its results do not establish suitability for medical, legal, or other consequential translation. Standard Spanish benchmark performance should not be generalized to local community or heritage varieties. LLM diagnostics can state unsupported errors with apparent confidence and should not be presented as expert explanations.

The public repository contains code, prompts, pinned configuration, reports, and numeric summaries. Raw benchmark text, full generations, vectors, model weights, credentials, and request identifiers are excluded from the public update. Access to underlying datasets and models remains subject to their respective terms. Norris artifacts remain separate from TRACE-MT; the earlier ICAI work is cited as prior methodology rather than merged into the experiment.

Draft prose and development analysis were prepared with AI assistance and checked against the supplied evidence. The project authors remain responsible for factual accuracy, attribution, and the venue's final disclosure requirements. Authorship, acknowledgments, funding wording, and publication metadata require PI confirmation before submission.

VIII. Limitations

The completed study is a 100-ID development pilot. Repeated model/strategy outputs are dependent, and paired directions share semantic content. Development intervals are descriptive after prompt development; they are not held-out confirmation. Two language pairs cannot identify a causal effect of language-resource availability. Baseline quality differs across directions, and some quality strata are sparse.

FLORES+ sources largely originate in English, so reverse directions involve translated source text and can exhibit translationese effects. This further limits a causal interpretation of directional asymmetry. Its independent target references are not minimal human post-edits, and neither editing effort nor user productivity is measured.

One open-weight checkpoint and one dated commercial snapshot do not represent all LLM families, scales, or contemporary models. Pretraining exposure to the public benchmark cannot be ruled out. Deterministic decoding and pinned model IDs improve traceability without guaranteeing identical remote behavior. Independent fragment translation can lose cross-sentence context.

COMET and heuristic flags are imperfect proxies; no independent bilingual annotation was collected. The source-based case review illustrates mechanisms rather than population error rates. The independent audit verifies numeric provenance and aggregation, not a fresh neural scoring run. The ICAI comparison uses its supplied final manuscript, without a byte-level audit of its unavailable code or data. Main statistical contrasts, multiplicity handling, and protocol freeze remain to be finalized before devtest evaluation.

IX. Development Conclusion and Remaining Evaluation

The completed pilot shows why post-editing evaluation must track preservation and operational validity alongside mean quality. GPT one-shot provides a positive English-to-Tagalog development estimate, while strict reason-then-edit validation exposes a substantial Llama compliance limitation. Case inspection shows that metric gains can accompany factual corruption and that large metric losses need not imply equally clear meaning loss.

These observations guide protocol review; they do not answer the four research questions conclusively. The next stages are to resolve the documented overlap and output-policy decisions, record the frozen configuration, execute the held-out matrix, and integrate the prespecified comparative analysis. Final results and conclusions will be revised from that evidence rather than projected from this pilot.

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Appendix A
Pinned Artifacts and Development Tables

The public companion is https://github.com/SHAZAN01/csudh-norris-ape-2026. The measured prompts are candidate v2, with their complete identifier, exact bytes, schemas, sample IDs, revisions, and hashes in the reviewed candidate JSON under `config/phase3/`. The current freeze flag is false. Historical "GPT pending" fields inside the pre-GPT preparation provenance are retained as historical bytes, not as the current execution status.

Revision prefixes are listed below for readability; the manifest retains the complete immutable identifiers.

Artifact	Revision prefix
FLORES+	`5fec6c13f9e5`
NLLB-200 3.3B	`1a07f7d19589`
LaBSE	`836121a0533e`
Llama model/tokenizer	`8afb486c1db2`
COMET	`2760a223ac95`

Retrieval used sentence-transformers 3.4.1 and FAISS CPU 1.9.0; it does not require a GPU FAISS build. Llama generation used vLLM 0.11.2, torch 2.9.0, and Transformers 4.57.1. Evaluation used torch 2.6.0+cu124, Transformers 4.49.0, unbabel-comet 2.2.7, NumPy 1.26.4, and SacreBLEU 2.6.0. Generation and evaluation are isolated environments; they should not be represented as a single tested installation.

The report's analysis signature begins `9d6830a0b9c7cc97`; the complete hash is recorded in the status manifest. Numeric audit artifacts contain all 24 conditions; the private original evidence contains raw completions and exact source/hypothesis/reference alignment. An audit script can reproduce lexical scoring from that evidence and check COMET aggregation without GPU inference.